

Power Query Transformations Guide

Lockheed Martin | Cleaning Financial Data in Power BI

Introduction

Real-world data from Dynamics 365 and Dataverse exports is never clean. This guide walks you through fixing common data quality issues using **only the Power Query Editor ribbon buttons** — no code required.

How to Open Power Query Editor

- 1. In Power BI Desktop, go to **Home > Transform Data**
- 2. The Power Query Editor window opens
- 3. Each table appears on the left under "Queries"
- 4. Your cleanup steps are recorded on the right panel under **Applied Steps**

Tip: Turn on **Column Quality**, **Column Distribution**, and **Column Profile** from the **View** tab. These show data issues at a glance.

File 1: Budget vs Actuals

What's Wrong With This Data?

Problem	What It Looks Like
Dollar signs in numbers	\$150000, -\$120000
Percentage signs	6.0%
Inconsistent capitalization	materials, travel, OVER BUDGET, at risk
Extra spaces	" Cyber Defense Initiative"
N/A text in number columns	N/A
Footer/notes rows at the bottom	"Notes: Data exported from..." + blank row + duplicate row
Shortened names	"Satellite Comm Upgrade" instead of full name
Multiple fiscal years & quarters	FY2025 (Q1–Q4) and FY2026 (Q1–Q2)

How to Fix It

1. Remove the footer/notes rows at the bottom

- **Home > Remove Rows > Remove Bottom Rows > Type 3**
- *(This removes the blank row, notes row, and duplicate row at the bottom in one step)*

2. Trim extra spaces from text

- Select all text columns (hold Ctrl and click: Program Name, Program ID, Cost Center, Budget Category, Funding Source, Status)
- **Transform > Format > Trim**

3. Fix dollar signs and commas in amount columns

- Select **Budgeted Amount**, **Actual Amount**, and **Variance** columns (hold Ctrl to multi-select)
- **Transform > Replace Values > Find: \$ Replace with: (leave blank) > OK**
- **Transform > Replace Values > Find: , Replace with: (leave blank) > OK**

4. Fix percentage signs

- Select **Variance %** column
- **Transform > Replace Values > Find: % Replace with: (leave blank) > OK**

5. Fix N/A values

- Select **Actual Amount**, **Variance**, and **Variance %** columns
- **Transform > Replace Values > Find: N/A Replace with: (leave blank) > OK**

6. Change columns to the right data type

- Click the "ABC" icon on **Budgeted Amount** column header > Select **Decimal Number**
- Click the "ABC" icon on **Actual Amount** column header > Select **Decimal Number**
- Click the "ABC" icon on **Variance** column header > Select **Decimal Number**
- Click the "ABC" icon on **Variance %** column header > Select **Decimal Number**

7. Fix inconsistent capitalization in Budget Category

- Select **Budget Category** column
- **Transform > Format > Capitalize Each Word**

8. Fix "Sub-Contracts" → "Subcontracts"

- Select **Budget Category** column
- **Transform > Replace Values > Find: Sub-Contracts Replace with: Subcontracts**

9. Fix Status column consistency

- Select **Status** column
- **Transform > Format > Capitalize Each Word**

10. Fix shortened program names

- Select **Program Name** column
- **Transform > Replace Values:**
 - Find: **Satellite Comm Upgrade** Replace with: **Satellite Communications Upgrade**

◦ Find: **Next Gen Radar Systems** Replace with: **Next-Gen Radar Systems**

File 2: Purchase Orders

What's Wrong With This Data?

Problem	What It Looks Like
Different date formats	10/15/2024, 11-20-2024, 2025-02-01, Feb 10 2025
Dollar signs and commas	"\$1,250,000.00"
NULL and dashes as values	NULL, -
Inconsistent status capitalization	CLOSED, closed, in progress
Inconsistent priority capitalization	CRITICAL, medium, high
Extra spaces in names	"Triton Systems ", " James Mitchell"
Vendor name variations	"Apex Dynamics Corp" vs "Apex Dynamics Corp."
Duplicate row	PO-1006 appears twice
Shortened program name	"Satellite Comm Upgrade"

How to Fix It

1. Remove duplicate rows

- Select the **PO ID** column > **Home** > **Remove Rows** > **Remove Duplicates**

2. Trim spaces from all text columns

- Select all text columns > **Transform** > **Format** > **Trim**

3. Clean the dollar amounts

- Select **PO Amount** column
- **Transform** > **Replace Values**: Find \$ Replace with blank
- **Transform** > **Replace Values**: Find , Replace with blank
- Repeat for **Amount Paid** and **Amount Remaining**

4. Fix NULL and dash values

- Select **Amount Paid** column
- **Transform** > **Replace Values**: Find NULL Replace with 0
- **Transform** > **Replace Values**: Find - Replace with 0

5. Change amount columns to Decimal Number type

- Click column type icon on **PO Amount** > **Decimal Number**

- Repeat for **Amount Paid** and **Amount Remaining**

6. Fix date columns

- Select **Order Date** column > Click the type icon > Choose **Date**
 - Power BI will auto-detect most formats
 - If any rows show "Error", right-click the column > **Replace Errors** > enter **null**
- Repeat for **Delivery Date**

7. Fix PO Status capitalization

- Select **PO Status** column
- **Transform** > **Format** > **Capitalize Each Word**

8. Fix Priority capitalization

- Select **Priority** column
- **Transform** > **Format** > **Capitalize Each Word**

9. Fix vendor name inconsistencies

- Select **Vendor Name** column
- **Transform** > **Replace Values**: Find **Apex Dynamics Corp.** Replace with **Apex Dynamics Corp**
- **Transform** > **Replace Values**: Find **Centurion Inc.** Replace with **Centurion Inc**

10. Fix shortened program names

- Select **Program Name** column
- **Transform** > **Replace Values**: Find **Satellite Comm Upgrade** Replace with **Satellite Communications Upgrade**

File 3: Programs (Dimension Table)

What's Wrong With This Data?

Nothing! This is a clean lookup table with one row per program. It serves as the central dimension for the star schema.

Steps

1. Verify data types

- All columns should import as **Text** — no changes needed

2. Rename the query

- Right-click the query name on the left > **Rename** > **Programs**
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Tips for the Exercises

How to Know You're Done Cleaning

Use the **View** tab and enable:

- **Column Quality** — shows % valid, error, and empty per column
- **Column Distribution** — shows distinct/unique value counts
- **Column Profile** — shows statistics for the selected column

You want:

- 0% errors in all columns
 - Number columns showing as numbers (not text)
 - Consistent values (check Column Distribution — you shouldn't see **Closed** AND **closed**)
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Common Mistakes to Avoid

1. **Don't forget to click "Close & Apply"** when done — otherwise nothing saves
 2. **Order matters** — Remove rows FIRST, then fix values (don't waste time cleaning rows you'll delete)
 3. **Watch for partial matches** — When replacing text, make sure you don't accidentally create doubled words
 4. **Check date conversions** — After changing to Date type, look for "Error" values that didn't convert
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Undo a Step

If you make a mistake:

- Look at **Applied Steps** on the right panel
 - Click the **X** next to any step to delete it
 - Steps below it will also be removed
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After Cleanup: Setting Up Relationships

Once all 3 tables are clean and loaded:

1. Go to **Model View** (left sidebar)
2. Drag **Program ID** from **Programs** to **Program ID** in **Budget vs Actuals**
3. Double-click the relationship line to open the dialog:
 - Cardinality: **One-to-Many**
 - Cross filter direction: **Single**
4. Drag **Program ID** from **Programs** to **Program ID** in **Purchase Orders**
5. Double-click the relationship line:
 - Cardinality: **One-to-Many**
 - Cross filter direction: **Single**

Why Single? In a star schema, filters flow from the dimension table (Programs) down to the fact tables. A slicer on Program Name will filter both Budget vs Actuals and Purchase Orders automatically. "Both" is

generally not recommended — it can cause ambiguity and performance issues.

The Programs table is your **dimension table** (star schema center). Both Budget vs Actuals and Purchase Orders are fact tables that connect to it. This enables cross-filtering between all tables through the shared Program dimension.

Next: See `REPORT_BUILDING_GUIDE.md` for how to build each report.